



Robust non-sticky/sticky antifouling coating capable of temporal evolution

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ABSTRACT

Antifouling coatings, which have important value for submerged marine structures, are faced with the challenge of inherent conflicting property requirements, i.e. the topcoat is desirable to prevent adherence of marine organisms and the basecoat should be able to maximize bonding to the substrate. Besides, the existing antifouling coatings have other tricky problems, like easy depletion of liquid lubricant of slippery liquid-infused porous surface and poor static microbes resistance of hydrophobic surface. Herein, a facile solution is proposed using the newly developed reversibly interlocked polymer networks, which enables temporally and spatially controllable multiple antifouling actions. Aided by dynamic covalent chemistry, the immiscible crosslinked polymers carrying various components with unique functionalities are uniformly interweaved producing a novel non-sticky/sticky antifouling composite coating (NSCC). Consequently, a series of non-contact and contact antifouling strategies based on physical and chemical interactions are seamlessly combined with high cohesion and interfacial adhesion. Either the surface or the subsurface of the NSCC exhibits excellent resistance to protein and bacteria under both static and dynamic conditions. The properties of the NSCC comprehensively outperform those of the analogues reported in the literatures. From a practical perspective, the redundancy design has set up a novel paradigm for preparing long-lasting antifouling coating.

1. Introduction

Painting antifouling coating onto submerged marine structures (like ship hulls, cross-sea bridges and seawater pipelines) is a critical measure for maritime-based industries [1–5]. This type of coating is a typical material that requires both excellent functionality (capable of preventing the attachment of marine organisms) and mechanical properties (i.e. strong cohesion of the coating bulk and adhesion to the substrates to withstand the impact of water flow). Obviously, such non-sticky and sticky performance requirements are conflicting, which are difficult to be integrated in the same material. The researches in this area mostly focus on the antifouling effect, paying less attention to the aforesaid mechanical properties that are of equal importance for the coating applications. As a result, the coating would be easy to fall off and suffer damage during service, and can no longer work properly.

As for the antifouling strategies of the coating's external layer, a series of techniques have been developed, e.g., release of embedded antifoulant [6], bioinspired micro/nanostructured surface [7],

hydrophilic surface [8], Zwitterionic coating [9], and amphiphilic surface [10]. Nevertheless, they are usually accompanied by the problems like complicated, inefficient and expensive fabrication, lack of broad-spectrum antifouling performance, poor antifouling durability and adhesion-to-substrate. Comparatively, the following three hold great promise for commercialization, i.e. degradable surface [11], low surface energy hydrophobic surface [12] and slippery liquid-infused porous surface (SLIPS) [13], and the latter two are easier to use because the coating can keep away from fouling organisms in principle, without worrying about their subsequent on-site growth. Still, however, biofouling may have a chance to settle down on the hydrophobic surface under static conditions, while SLIPS may lose its ability upon depletion of the pre-embedded lubricant.

Herein, we put forward a new concept antifouling coating, which not only combines the advantages of the aforesaid methods to resist biofouling and overcomes the drawbacks, but also improve the mechanical properties that used to be neglected. The core advancement of the design is based on the reversibly interlocked polymer networks

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(RILNs), which are obtained via topological rearrangement of two crosslinked polymers containing orthogonal reversible covalent bonds [14–16]. Owing to the forced miscibility of RILNs, polymer chains of different properties can be interweaved at the molecular level regardless of their miscibility, which is unavailable for the traditional interpenetrating polymer networks (IPNs). As a result, the antifouling coating prepared according to the RILNs method is planned to consist of one subnetwork responsible for hindering biofouling and the other for bonding the substrate. The interlocking architecture formed by the two polymer networks ensures the coating's cohesion.

For verification, hydrophobic and hydrophilic crosslinked polyurethanes are synthesized first (Fig. 1a and 1b), respectively. The former (called Schiff-PU, refer to Figs. S1 and S2) employs double amino-terminated polydimethylsiloxane (PDMS) as soft segment and reversible Schiff base bond as antimicrobial chain extender [17], while the later (denoted by DA-PU, refer to Figs. S3 and S4) employs poly (butyleneadipate-co-terephthalate) (PBA) as soft segment and reversible Diels-Alder (DA) bond as crosslinker. In addition, the environmentally friendly ceftazidime antibacterial monomer (CAZ) is grafted to the hard segment of Schiff-PU for sterilization. The selection of CAZ as the

antibacterial monomer can be attributed to three key factors. Firstly, CAZ is a green and environmentally friendly antibacterial agent that is primarily utilized in the field of biomedicine and can be taken orally. Secondly, CAZ has been demonstrated to possess a broad-spectrum antibacterial activity [18], showing notable efficacy against both gram-positive and gram-negative bacteria. Notably, it has not yet been applied to antifouling coatings, therefore, there is no risk of drug resistance. Finally, the amino group present in CAZ monomer facilitates its grafting to the polymer.

Afterwards, following the approach of creating RILNs [14–16], Schiff-PU is mixed with DA-PU under heating in their co-solvent by taking advantage of disconnection of the built-in reversible bonds (Figs. 1c, S5 and S6), and then the mixture is coated onto a piece of epoxy sheet that simulates the common primer of marine steel (Fig. 1d). During the subsequent evaporation of the co-solvent and cooling, fragments of the temporarily de-crosslinked DA-PU in the above mixture are enriched near the boundary with the epoxy sheet owing to their similar surface energies, spreading out on the sheet and filling in the surface grooves (Fig. 1e). As a result, fragments of Schiff-PU are naturally enriched at the outward side of the coating at the same time.

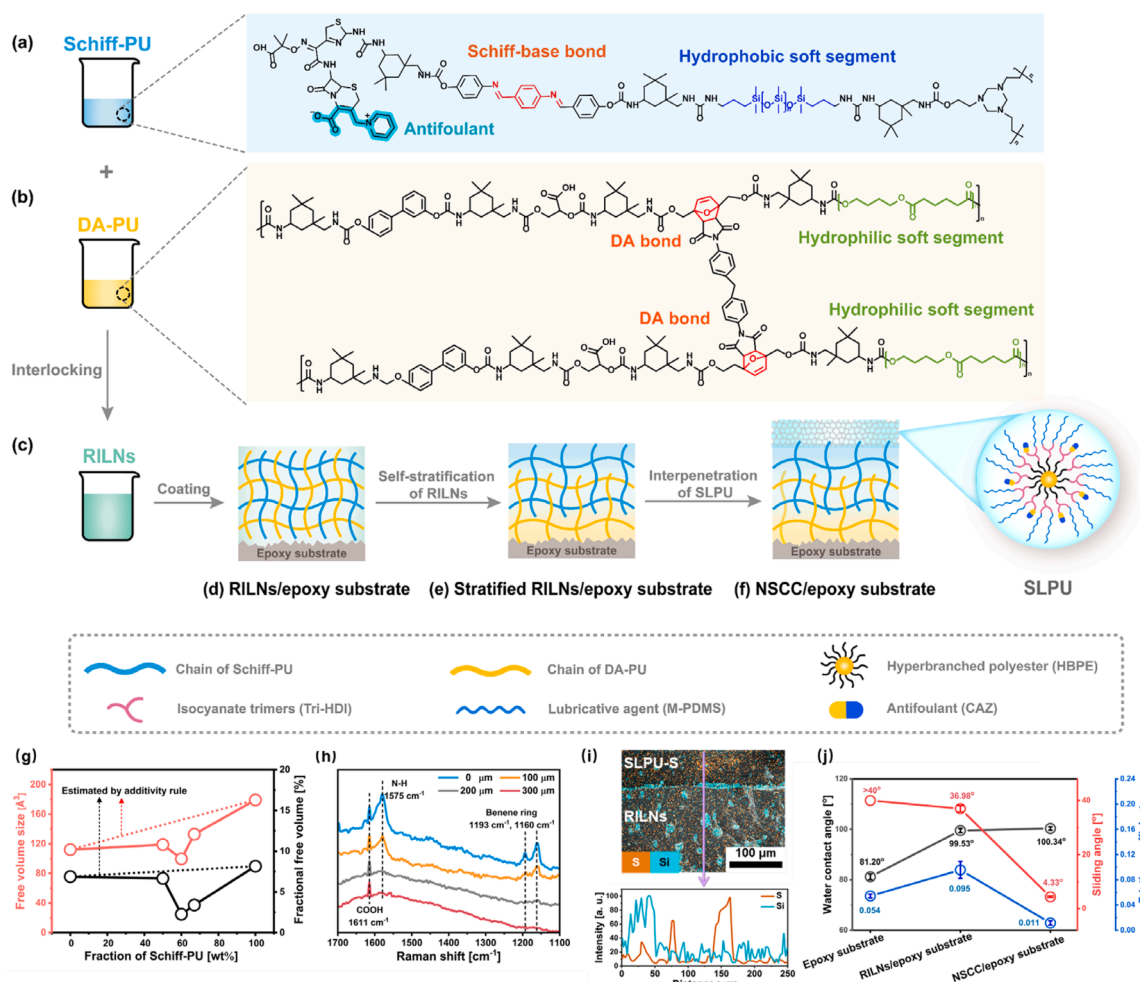


Fig. 1. Design and structure of non-sticky/sticky antifouling composite coatings (NSCC). (a and b) Chemical structures of (a) Schiff-PU and (b) DA-PU. (c-f) Preparation processes of NSCC, including (c) preparation of RILNs, (d) coating of RILNs onto epoxy substrate, and (e) self-stratification of the RILNs coated onto epoxy substrate, and (f) interpenetration between SLPU and stratified RILNs coated onto epoxy substrate. (g) Average free volume hole size and fractional free volume as a function of the Schiff-PU content in the RILNs acquired from positron annihilation lifetime measurement. (h) Raman depth scanning spectra of the RILNs (which was peeled from epoxy substrate (e)) measured in the direction from the top side (away from the epoxy substrate) to the down side (near the epoxy substrate). (i) Energy dispersive spectrum (EDS) mapping and the corresponding line-scanning curves of S and Si of the partial cross-section of the NSCC. Here, the surface SLPU has been labelled by S in advance for identification, and the line-scanning curves were measured from the S-labelled SLPU (i.e. SLPU-S) part towards RILNs. The SLPU-S/RILNs sample was peeled from the epoxy substrate. (j) Water contact angles, sliding angles and friction coefficients of surfaces of epoxy substrate, RILNs coated epoxy substrate, and NSCC coated epoxy substrate.

Under the circumstances, the prepolymer solution of the third polyurethane (i.e., solid-state self-lubricating polyurethane (SLPU), refer to Figs. 1f, S7a and S8), in which monofunctional PDMS (M-PDMS) and CAZ are attached to the hard segments as floating solid-state lubricant and antifoulant, respectively, is overlaid and cured. As temperature drops further, hydrogen bonds at the DA-PU/epoxy substrate interface are established forming robust adhesion, and reconnection of the reversible bonds leads to generation of RILNs from the entangled DA-PU and Schiff-PU fragments, which enhances cohesion of the coating. Besides, the molecules of SLPU are rooted into the reformed Schiff-PU producing a tightly attached slippery hydrophobic layer.

It is anticipated that under the guidance of the above holistic improvement philosophy, our non-sticky/sticky antifouling composite coating (NSCC) would be rather mechanically robust as a result of the high cohesion and strong affinity for the substrate, and in particular, possess superior long-lasting static and dynamic antifouling properties owing to the environmentally adaptively temporal evolution of various fouling resisting mechanisms implanted in different locations.

2. Material and methods

2.1. Materials

Hexamethylene diisocyanate trimer (HDIT) was purchased from Bayer AG. Hydroxyl-terminated hyperbranched polyester (HBPE) was supplied by Wuhan Hyperbranched Polymer Science & Technology Co., Ltd. Mono-hydroxyl-terminated polydimethylsiloxane (M-PDMS, $M_n = 3000 \text{ g mol}^{-1}$) was bought from Foshan Silchem Trading Co., Ltd. Poly (butyleneadipate-co-terephthalate) (PBA, $M_n = 1000 \text{ g mol}^{-1}$), ceftazidime (CAZ), 4-aminothiophenol (ATP), and dibutyltin dilaurate (DBTDL) were purchased from Shanghai Maclin Biochemical Technology Co., Ltd. PBA was dehydrated at 110°C in vacuum for $>6 \text{ h}$. Double amino-terminated PDMS ($M_n = 3000 \text{ g mol}^{-1}$), *p*-phenylenediamine (PPD), 4-hydroxybenzaldehyde (HBD), 4,4'-dihydroxybiphenyl (DP), glyceric acid (GA), and 2,5-furandimethanol (FA) were supplied by Innocem Technology Co., Ltd. Isophorone diisocyanate (IPDI), hexahydro-1,3,5-tris(hydroxyethyl)-s-triazine (ACT), bismaleimide (BMI), acetic acid, and deuterated chloroform (CDCl_3) were purchased from Anhui Zesheng Technology Co., Ltd. Anhydrous ethanol, ethyl acetate, diiodomethane, anhydrous tetrahydrofuran (THF) and *N,N*-dimethylformamide (DMF) were bought from Innocem Technology Co., Ltd. Deionized water (DI water) was purchased from A.S. Watson Group (Hong Kong) Ltd. Glutaraldehyde was obtained from Sigma Aldrich Chemicals Pvt. Ltd. Fluorescein isothiocyanate (FITC) marked bovine serum albumin (BSA) solution (FITC-BSA) was purchased from Beijing Solarbio Science & Technology Co., Ltd. Phosphate-buffered saline (PBS) was purchased from Grand Island Biological Company. *Pseudomonas aeruginosa* (*P. aeruginosa*) was supplied by Guangyu Biological Technology Co., Ltd. Luria-Bertani (LB) plates, liquid medium and solid medium were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. The artificial sea salt was purchased from Guangzhou Yier Biological Engineering Co., Ltd. All chemicals and solvents were used as received without further purification unless otherwise stated.

The epoxy plate (80 mm long, 10 mm wide and 2 mm thick) was purchased from Guangdong Hongwang New Material Technology Co., Ltd. The sandpapers (grit numbers: 100, 400, 800, 1200 and 2500) were purchased from Beijing Lanjiek Technology Co., Ltd. The epoxy plates used in the current work were mostly pre-polished with 100 grit sandpaper unless otherwise specified. The 304 McMaster-Carr stainless steel wire cloth (mesh size: 170×10) was supplied by Suzhou Linzhe Hardware Product Co., Ltd. The glass slide (75 mm long, 10 mm wide and 1 mm thick) was purchased from CITOTEST Labware Manufacturing Co., Ltd.

2.2. Synthesis of hydrophobic polyurethane network (Schiff-PU, i.e. crosslinked polyurethane containing reversible Schiff-base bonds)

Firstly, the hydroxy terminal phenylene core aromatic Schiff-base monomer (Schiff-base monomer) was synthesized according to the method reported elsewhere (Fig. S1a) [19]. PPD (4.325 g) and HBD (10.258 g) were dissolved in 150 mL anhydrous ethanol, respectively. Afterwards, the HBD solution was dropped into the PPD solution within 3 h, and the mixture was stirred for 8 h at room temperature. Having been filtered, the solid product was cleaned by filtration and washed by ethyl acetate for several times. The resultant yellow solid was collected and dried at 45°C for 48 h.

The synthesis procedures of Schiff-PU are shown in Fig. S1b-S1d. Typically, double amino-terminated PDMS (15 g) was dissolved in anhydrous THF (20 mL), and then IPDI (2.45 g) was dropped into the solution. Having been stirred at room temperature for 12 h under argon protection, the prepared Schiff-base monomer (1.58 g) and DBTDL (three drops) were added and the system was heated to 60°C for 12 h for polymerization. Next, the THF solution (5 mL) of ACT (0.11 g) was incorporated into the mixture for additional 12 h. Finally, CAZ (0.27 g) was added under intensely stirring for a few minutes, and the resultant solution was poured into a polytetrafluoroethylene (PTFE) mold for curing at 60°C for 48 h, yielding orange colored polymer film of Schiff-PU.

2.3. Synthesis of hydrophilic polyurethane network (DA-PU, i.e. crosslinked polyurethane containing reversible Diels-Alder bonds)

The synthesis of DA-PU is similar to that of Schiff-PU, which contains pre-polymerization, chain-extending and cross-linking (Fig. S3). First, the dried PBA (10 g) was mixed with IPDI (5.78 g) and DBTDL (four drops) for pre-polymerization, and then different chain-extenders (GA (0.268 g), DP (1.117 g), and FA (1.025 g)) were added. Next, BMI monomer (1.443 g) was introduced to the system for crosslinking the polymeric chains. All the reactions took place at 60°C and the time for each step was 12 h. Lastly, the mixture was poured into a PTFE mold and cured at 60°C for 48 h in vacuum to obtain a colorless transparent film of DA-PU.

2.4. Preparation of the reversibly interlocked polymer networks (RILNs)

According to the method of preparing RILNs reported in our early works [14–16], DMF was selected to be the co-solvent for dissolving the pre-made single networks (i.e. Schiff-PU and DA-PU), and the preparation temperature was set at 100°C because the retro-DA reaction temperature is predominantly within the range from 100 to 120°C [20–22], and the exchange reaction temperature of Schiff-base is between room temperature and 190°C [23–26].

Specifically, the Schiff-PU (5 g) and DA-PU films (5 g) were cut into small pieces and soaked in DMF (100 mL) at room temperature for 24 h to get full swelling, respectively. Then, the gel of swollen Schiff-PU was heated to 80°C for 1 h, and that of DA-PU was heated to 100°C for 10 min under the protection of argon. Lastly, the two solutions were mixed under different proportions at 100°C for 1 h. The resulting homogeneous solution was filtered and poured into a PTFE mold for drying at 60°C for 48 h. The product was named as RILNs-XY, which means that the weight ratio of Schiff-PU to DA-PU is X:Y. RILNs-32 and RILNs-42 were prepared for purposes of verifying the versatility of the interlocking structure and the leading RILNs studied in the current work is RILNs-22, so that it is simply referred to as RILNs unless otherwise specified. The reason behind this decision is attributed to the fact that our previous research [27] demonstrated that the interlocking structure is the most homogeneous when the two subnetworks are mixed in equal proportions. The enhanced interchain interactions between the two subnetworks should be responsible for the above-mentioned phenomenon under the circumstances.

2.5. Preparation of the control of the RILNs (i.e. simultaneous interpenetrating polymer networks, IPNs)

To highlight the significance of RILNs, simultaneous IPNs were prepared from the same raw materials for making RILNs. Typically, PDMS (15.19 g), PBA (10 g), IPDI (8.25 g), Schiff-base monomer (1.60 g), ACT (0.11 g), CAZ (0.27 g), GA (0.268 g), DP (1.117 g), FA (1.025 g), BMI monomer (1.443 g), and DBTDL (seven drops) were dissolved in a mixed solution of DMF (50 mL) and THF (50 mL). The polymerization was carried out at 60 °C for 24 h under stirring in dry argon. Then, the product was poured into a PTFE mold and cured at 60 °C for 48 h.

2.6. Preparation of solid-state self-lubricating polyurethane (SLPU) and its sulfur labelled version (SLPU-S)

To prepare SLPU, HBPE (2 g), HDIT (4.52 g), different weight contents of M-PDMS (0, 0.25, 0.50, 1.00 and 2.00 wt%) and CAZ monomer (0.5, 1.0 and 1.5 wt%) were mixed and then DMF was added until full dissolution. Thereafter, DBTDL (40 μ L) was incorporated, and the mixture was rapidly cast onto the epoxy substrate and cured in an oven at 60 °C for 12 h (Fig. S7a).

2.7. Preparation of non-sticky/sticky composite coating (NSCC)

As shown in Fig. 1c–1f, the preparation process of NSCC is divided into the following steps. The epoxy sheet was polished by 100 grit sandpaper, followed by washing with deionized water and anhydrous ethanol for 1 h each and dried. Afterwards, RILNs were painted on the polished epoxy sheet serving as the substrate (Fig. 1d–1e). Finally, SLPU was coated on the cured RILNs (Fig. 1f). Specifically, the resulting homogeneous RILNs solution (refer to sub-Section 2.3 of the Supporting Information for more details) was concentrated into 1 g mL⁻¹, and then the condensation was dropped on the epoxy substrate. The epoxy substrate coated with RILNs solution was cured at 60 °C for 12 h (coating thickness $\approx$ 300–400 μ m). Next, the solution of SLPU was dropped on the cured RILNs coating (refer to sub-Section 2.5 of the Supporting Information for more details) and dried at 60 °C for 2 h to form a 100 μ m thick lubricating shell.

2.8. Characterization

Proton nuclear magnetic resonance (¹H NMR) spectra were recorded with an AVANCE III NMR spectrometer (400 MHz) at 298 K by using deuterated chloroform (CDCl₃) as the solvent and tetramethylsilane as the internal standard. The chemical shifts were referenced to the peak of CDCl₃ (7.26 ppm).

Attenuated total reflection Fourier transform infrared (ATR-FTIR) spectra were collected within the wavenumber region of 4000–400 cm⁻¹ by a Bruker EQUINOX55 spectrometer at a spectral resolution of 4 cm⁻¹ with 64 scans.

Ultraviolet-visible (UV–vis) absorption spectra were tested at room temperature in the wavelength range of 180–800 nm by means of a Perkin Elmer Lambda 950 spectrometer. The measurements were conducted in two modes. In the case of liquid samples, transmission mode was employed, while in the case of solid films, integrating sphere mode was utilized.

Florescence emission spectra were measured by a Hitachi F-4500 florescence spectrometer. The excitation wavelengths were 300 nm, 350 nm and 400 nm, respectively. The scan speed was 1200 nm min⁻¹ and the photomultiplier tube (PMT) voltage was 700 V.

Raman spectra were measured using single spot mode with a Thermo Fisher DXR3xi confocal Raman microscope (wavelength of the excitation laser = 785 nm, laser intensity = 30 mW, exposure time = 0.1 s, and accumulation times = 1000).

Differential scanning calorimetry (DSC) tests were carried out using a TA Instrument DSC Q10 at a heating rate of 20 °C min⁻¹ in nitrogen

atmosphere. The samples were firstly heated up to 100 °C, kept for 5 min to eliminate the thermal history, and then quenched to –80 °C. Next, the samples were heated from –80 °C to 100 °C. The data of the last heating process were collected and analyzed by TA Instruments universal analysis software.

Dynamic mechanical analysis (DMA) was conducted on a TA-DMA 8500 using tension mode under 1 Hz at a heating rate of 3 °C min⁻¹. The temperature sweeps were carried out under a strain amplitude of 1 % within –60 °C and 100 °C. The samples were firstly cooled to –60 °C, kept for 5 min, and then heated to 100 °C. The cooling or heating rate of each step was 3 °C min⁻¹. The storage modulus (G') and loss modulus (G'') of the specimens were collected during the heating process and the loss factors (tan δ) were calculated.

The microstructure and element distribution of the samples were examined by using a ZEISS Gemini SEM250 thermal field-emission scanning electron microscope (SEM) equipped with an energy dispersive spectroscopy (EDS) detector. Before the measurements, the sample was quenched in liquid nitrogen for 10 min and broken to expose the cross-section. Later, the cross-section was coated by Au sputtering. Besides, DSX10-SZH digital microscope was employed to observe the macroscopic morphology. Atomic force microscope (AFM, Bruker, Multimode 8) was applied to study surface topography and modulus via the contact mode and the peak force quantitative nano-mechanical (QNM) mode at 25 °C, respectively. Prior to the QNM testing, the samples were dissolved in DMF and dropped onto a piece of silicon plate to obtain smooth surface.

Fractional free volume and the pore radius were tested using a positron annihilation lifetime spectrometer (ORTEC 0107A) with ²²Na as the radiation source. The radiation source was sandwiched between two samples and the lifetime spectra were recorded at room temperature until more than one million accumulated counts were achieved. Lastly, the resulting lifetime spectra were analyzed and calculated [14].

Water contact angle was measured at 25 °C by a DSA-100 contact angle analyzer (OCA50EC, Japan) using deionized water (3 μ L). Five different positions were tested and the average value was obtained. Surface energy was estimated by the Owens-Wendt-Rabel-Kaelble method using the contact angles of deionized water and diiodomethane determined at 25 °C. The sliding angle, which represents the inclination angle of the surface when a certain volume of deionized water is going to roll down, was also measured [28] by the above DSA-100 contact angle analyzer. Firstly, the specimen was fixed on the test bench, and then 25 μ L deionized water was dropped onto the specimen surface. Afterwards, the specimen was tilted until 40° that was sufficient for meeting the testing requirement, and the entire process was video recorded. The sliding angle was determined according to the initial displacement of the deionized water droplet. At least three different regions were tested and the average value was determined to the final result.

Frictional coefficient was determined by a MS-MX-D slider-on-disk detector [26,27]. The coated epoxy substrate was cut into 6.3 cm $\times$ 6.3 cm square, adhered to the stainless-steel slider (weight = 200 g), and immersed in artificial seawater (pH = 8.2–8.4) to simulate the under-water friction. The frictional coefficient between the coating and artificial seawater during uniform linear motion of the slider (speed = 100 mm min⁻¹) was recorded by a force sensor. The averaged results of four samples were considered as the final coefficient of friction.

For measuring the anti-protein performance, the samples (i.e. NSCC/epoxy substrate assemblies) were immersed in 2 mL FITC marked bovine serum albumin solution (FITC-BSA solution, concentration = 1 mg mL⁻¹) and incubated at 25 °C for 2 h in the dark. Afterwards, the samples were brought out and slightly rinsed by PBS solution to remove the unattached FITC-BSA. Finally, the samples surfaces were observed by a laser confocal microscope (LSM 710 NLO, Zeiss, Germany). Because FITC served as the fluorescent dye, an argon ion laser was employed for the observation.

The static antibacterial property was evaluated by the Luria-Bertani

(LB) plate method (Fig. 2a1), using *Pseudomonas aeruginosa* (*P. aeruginosa*, Gram-negative) as the representative bacteria [29]. Firstly, the bacteria were streaked on LB plates and allowed for growing at 37 °C for 24 h. Next, a monoclonal colony was inoculated into a tube containing liquid medium and cultured at 170 rpm at 37 °C for 12 h. The cultured bacterial solution was diluted into different multiples with PBS and assayed, yielding a suspension with a bacterial density of 1×10^8 CFU mL⁻¹. Meanwhile, all the samples were sterilized with a UV lamp (20 W, 253.7 nm) for 30 min and then immersed in 2 mL diluted bacterial solution (1×10^8 CFU mL⁻¹) and incubated at 37 °C for 24 h. Subsequently, 0.1 mL of the solution was taken out from the tube and diluted by 10^3 times with PBS and 100 μ L of the diluted solution was evenly scraped onto the LB solid medium and incubated at 37 °C for 24 h. The number of the bacterial colonies on the plates were determined by plate counting method, and the anti-bacterial rate was calculated from:

$$\text{Anti-bacterial rate (\%)} = (A - B)/A \times 100\%$$

where *A* and *B* are the colony numbers of the control group (i.e. the blank PBS) and the sample group, respectively.

The dynamic antibacterial experiment was performed under stirring to simulate the seawater washout environment of real-world applications. The blank glass slide, the glass slide coated with RILNs and the glass slide coated with NSCC/RILNs were adhered to a beaker (diameter = 65 mm), respectively. Then, 200 mL artificial seawater (pH = 8.2–8.4) containing 5 mL bacterial solution with a concentration of 1×10^8 CFU mL⁻¹ was filled in the beaker and washed with constant agitation at 1500 rpm (simulating a moderate sailing speed of 20 knots, Fig. 2a2). After 1, 3, 5 and 7 days of flushing, the samples were removed and rinsed with 2 mL artificial seawater. Next, the glass slides were cut into observable sizes, fixed by 2.5 % glutaraldehyde solution, dried at room temperature and inspected by SEM.

Tensile properties of dumbbell-shaped specimens (length = 50 mm, width = 4 mm and thickness = 2 mm) were measured at ambient

temperature by a WD-5A universal tester under a crosshead speed of 100 mm min⁻¹. The hardness was determined by the means of pencil test according to ASTM D3363–22 standard.

To understand the adhesion between the RILNs and epoxy substrate, lap shear and 180° peeling tests were conducted on the same universal tester, respectively.

In the case of lap shear test, the DMF solution of RILNs (1 g mL⁻¹) was dropped onto two pieces of the epoxy sheets (which were polished by 100 grit sandpaper in advance) and the samples were dried in an oven. Then, the RILNs coated epoxy sheets (80 mm × 10 mm × 2 mm) were overlapped forming a sandwich, in which the middle layer was RILNs. Afterwards, the jointed sheets were clamped and heated at 120 °C for 1 min to ensure adequate wetting between the RILNs and epoxy sheets. Having been cooled down to room temperature, the sample was tested to failure at a crosshead speed of 100 mm min⁻¹. The maximum force was recorded and divided by the bonding area (100 mm²) to obtain the final lap shear strength. At least three samples were tested and the average value was taken as the final result.

As for the 180° peeling test, the RILNs solution was dropped onto a piece of the polished epoxy sheet (80 mm × 10 mm × 2 mm, polished by 100 grit sandpaper in advance) and heated until semi-dry. Later, a stainless steel mesh (170 mm × 10 mm × 1 mm) was placed on the RILNs layer, and additional RILNs solution was poured onto partial stainless steel mesh. The length and width of the covered area were 60 mm and 10 mm, respectively. Along with the curing of the RILNs, the stainless steel mesh was firmly embedded in the RILNs and can thus act as a backplane for peeling [30]. Lastly, the free ends of the epoxy substrate and stainless steel mesh, which were not glued together, were attached to the universal tester and pulled to failure at an angle of 180° under the crosshead speed of 100 mm min⁻¹. The maximum force divided by the width of the stainless steel mesh told the peeling strength. At least three samples were tested and the average value was taken as the final result.

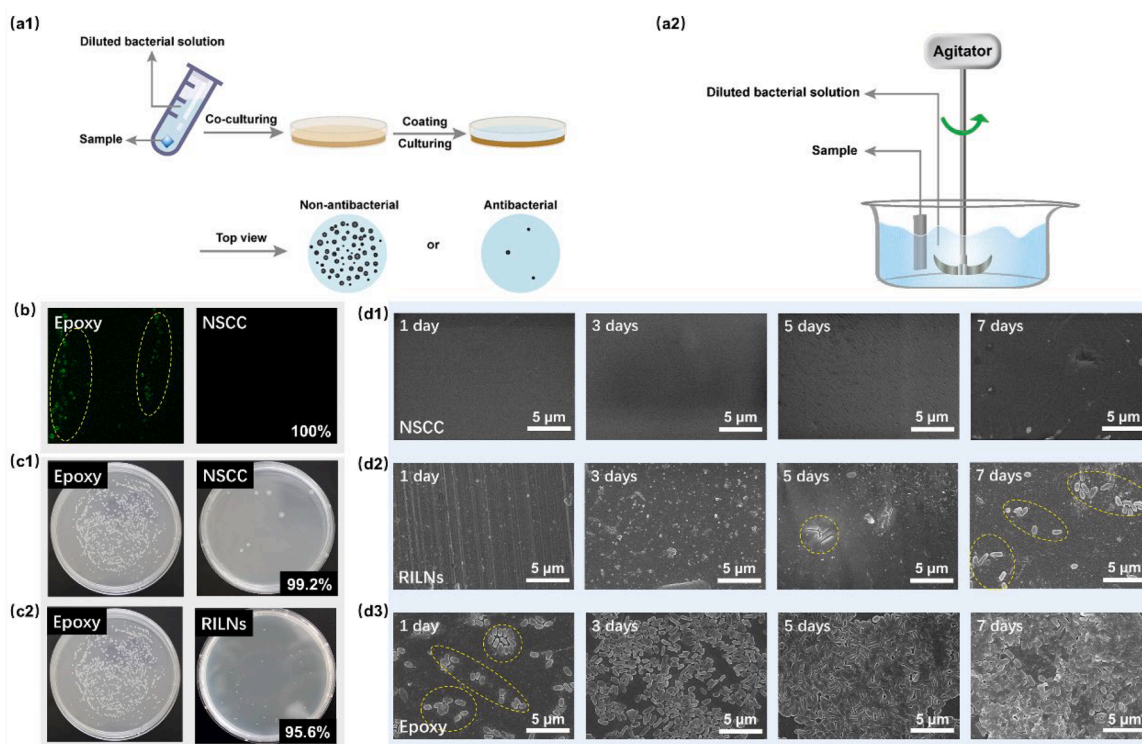


Fig. 2. Antibiofouling performance. (a1 and a2) Schematic illustrations of (a1) static anti-bacterial and (a2) dynamic anti-bacterial tests. (b) Fluorescence microscopy images of proteins adhered on bare glass and NSCC coated glass. (c1 and c2) Static anti-bacterial performance of NSCC (c1) and RILNs (c2) characterized via plate culture method. (d1, d2 and d3) SEM images of *P. aeruginosa* after 1, 3, 5, and 7 days of dynamic flushing on (d1) NSCC/epoxy substrate, (d2) RILNs/epoxy substrate, and (d3) blank epoxy substrate. Note: The dotted circles in (b, d2 and d3) highlight the attached protein and bacterial.

To simulate the self-degradation due to acid-responsive degradation of the built-in Schiff-base bonds, the RILNs coated epoxy substrate was immersed in acidic water (pH = 3.0) for different times, which not only imitated the acidic microenvironment caused by the aggregation, growth and metabolism of bacteria, but also speeded up the experiments [31]. Subsequently, surface morphology of the RILNs was monitored by a digital microscope and the corresponding roughness was measured. Besides, the variations in the surface chemical components were characterized by Raman spectroscopy, and the water contact angles and the surface energies were tested to examine the changes in surface properties after degradation.

3. Results and discussion

3.1. Fundamental structure-properties characterization

The successful synthesis of Schiff-PU and DA-PU (Fig. 1a and 1b) is confirmed by FTIR and Raman spectroscopy (Fig. S2b, S2c and S4). Then, the uniform RILNs (Fig. 1c) consisting of Schiff-PU and DA-PU also prove to have been made, according to the observations in terms of DSC, DMA, AFM and positron annihilation lifetime spectrometer (Fig. S5 and S6). In particular, both average free volume hole size and fractional free volume of the RILNs are lower than the linear additive law over a wide range of the content of Schiff-PU (Fig. 1g), manifesting that the macromolecules from the two subnetworks are interlaced with each other as expected.

Meanwhile, SLPU (Fig. S7a) has also been yielded as verified by the FTIR and Raman spectroscopy (Fig. S8a and S8b). When it is painted on the RILNs forming NSCC, the intensities of the characteristic infrared peaks of M-PDMS segment on the top side of SLPU are stronger than the average represented by the powdered SLPU (Fig. S8c and S8d). Unlike the bifunctional PDMS that is fixed in Schiff-PU, quite a few soft M-PDMS chains with only one ends bonded to the solid segments of SLPU must be able to migrate to the surface along with the solvent evaporation [32], which reduce surface energy of NSCC, endowing the latter with enhanced lubricating effect.

Fig. 1h shows the Raman depth scanning spectra of the RILNs separated from the epoxy substrate. The peak intensity of -NH groups that are unique to Schiff-PU increases with increasing distance to the bottom (i.e. the epoxy substrate side), while that of -COOH of DA-PU shows the opposite trend of variation (Fig. S9). It means that the envisaged self-stratification of RILNs (Fig. 1e) indeed happens, as driven by the differences in wettability of Schiff-PU and DA-PU to epoxy substrate (Fig. S10). Besides, the commingling between the Schiff-PU rich part of the RILNs and SLPU is also favored, as they have basically similar chemical compositions (Figs. S1 and S7a). To evidence the analysis, ATP is grafted onto SLPU in advance (Fig. S7b), so that sulfur can serve as a tracer of SLPU, distinguishing SLPU from the Schiff-PU rich part. The EDS mapping of the cross-section of the labelled SLPU covered RILNs and the corresponding line-scanning prove that S (i.e. SLPU) is not only densely distributed in the former but also diffuses into the RILNs (Fig. 1i).

In fact, the EDS results given in Fig. 1i also reveal the variation in concentration of Si in the two adjacent layers. The amount of Si is increased at the interface due to the superposition of infiltrated SLPU. It suggests that despite the long-chain M-PDMS in SLPU that has migrated to the surface for creating the solid lubricating effect, there is still enough M-PDMS at the bottom of SLPU to form strong interaction with the Schiff-PU rich part of the RILNs, as a result of the hyperbranched structure (Fig. S7a).

Since hydrophobic PDMS is present in Schiff-PU and SLPU, the water contact angles of the stratified RILNs/epoxy substrate without and with SLPU on the top layer become 99.53° and 100.34° (Fig. 1j), respectively, which are obviously greater than that of the epoxy (81.20°) and reflect that the coating surfaces have turned the hydrophilicity of the epoxy substrate into hydrophobicity. Furthermore, because one end of the M-

PDMS in SLPU is free to move serving as surface lubricant, which is confirmed by the spectroscopic measurements (Fig. S8c and S8d), the stratified RILNs with SLPU coverage (i.e. NSCC) has rather low sliding angle of water and frictional coefficient in water, which forms sharp contrast to the cases of the stratified RILNs without SLPU coverage and epoxy substrate (Figs. 1j and S11).

Considering the fact that the sliding behavior of SLPU is primarily affected by the content of M-PDMS, a series of SLPU with various quantities of M-PDMS are prepared and tested to identify the optimal dosage. It is seen from Fig. S12a and S12b that with the addition of M-PDMS, the water contact angle of SLPU increases significantly as the surface turns from hydrophilic to hydrophobic, while the sliding angle decreases dramatically. On the other hand, the antifoulant CAZ introduced into Schiff-PU and SLPU (Figs. S1 and S7a) has little influence on the water contact angles but slightly increases the sliding angle of water of SLPU (Fig. S12c and S12d). The polarity of CAZ might take the responsibility as its attraction to water would somewhat increase the hydraulic resistance. By taking these results into account, a formula consisting of 2 wt% M-PDMS and 1.5 wt% CAZ is regarded as the optimal combination for the preparation of SLPU to be used in the subsequent investigation.

3.2. Antifouling performance

As the RILNs based NSCC has been acquired, we can start to assess the antifouling performance by employing proteins and bacteria as the representatives of marine life. From a practical scenario, protein absorption is the first stage of marine biofilm formation, which provides the nutritional conditions for subsequent micro-adhesion of bacteria [33]. Therefore, it is discussed first.

Having been immersed in the solution of model protein (BSA) and incubated under static conditions, the samples are taken out for observation by laser confocal microscopy. The method is similar to that applied for assessment of static anti-bacterial performance (Fig. 2a1), except that the plate culturing after immersing is excluded. As shown in Fig. 2b, the fluorescent signals are clearly discernible on the blank glass, whereas the surface of NSCC exhibits no fluorescence. Evidently, the NSCC possesses excellent static anti-protein adhesion property (i.e. 100 % antiprotein rate). Meantime, when the incubation is conducted under dynamic conditions (Fig. 2a2) by using flowing model protein solution, still no attached BSA can be found on NSCC. This is understandable, because the antifouling process is more challenging in static conditions than that in dynamic conditions.

Besides the contamination of protein, we need to check whether bacteria can grow on the coating. As once proteins are anchored on the surface, bacteria would immediately follow and become the most significant and difficult part of biofouling. *Pseudomonas aeruginosa* (*P. aeruginosa*) is one of the most common bacterial species in marine and therefore acts as the representative bacterium for the antibacterial experiments. Moreover, the different bacterial adherences in the stationary state and the sailing state usually result in disparate antibacterial performances, so that static co-culture and dynamic scouring simulation methods are used to investigate the static and dynamic antibacterial properties, respectively.

Fig. 2c1 shows that after static co-cultivation of the samples and bacterial solution (Fig. 2a1), the plate of the blank sample exhibits a dense population of colonies, and that of NSCC only a sparse population of individual colonies. The static antibacterial resistance rate is estimated to be as high as 99.2 %. Fig. 2d1 illustrate the morphological changes in NSCC after the flushing of artificial seawater containing the bacterium for 1, 3, 5 and 7 days. Although the number of the bacteria adhering to the blank substrate rapidly increase over time (Fig. 2d3), no bacterial are found to be attached to the NSCC during the entire experimental period. The results indicate that the NSCC has excellent dynamic antibacterial property as characterized by 100 % antibacterial resistance. In this context, it can be concluded that the antifouling effect

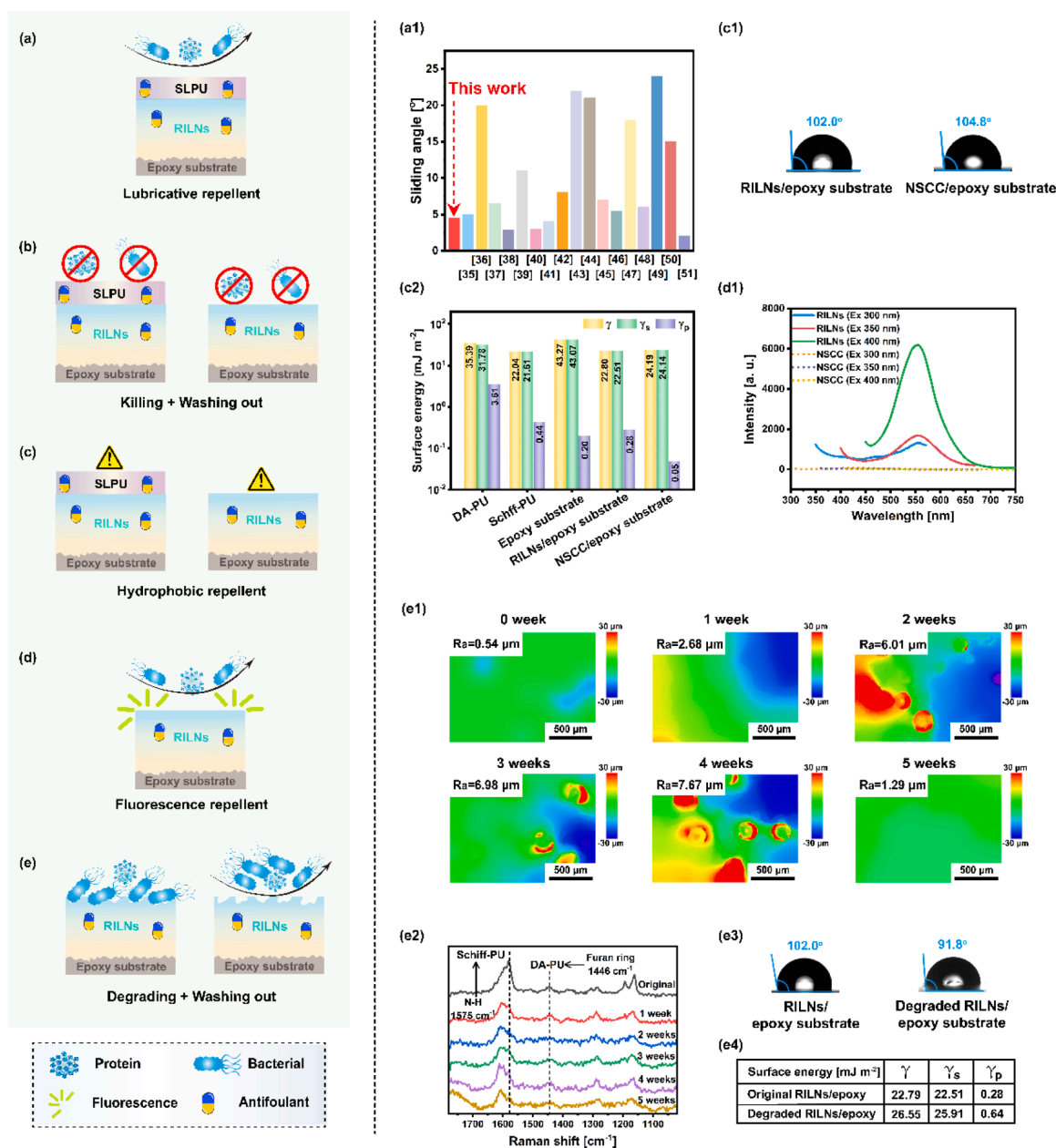


Fig. 3. Studies of antifouling mechanisms. (a-e) Schematic drawings of (a) lubricative repellent effect of NSCC, (b) fouling killing effect of NSCC and RILNs coating caused by the grafted bactericides (Fig. S1 and S7a), (c) hydrophobic repellent effect of NSCC and RILNs coating, (d) fluorescence repellent effect of RILNs coating, and (e) degradation induced renewal effect of RILNs coating. (a1) Sliding angle of NSCC/epoxy substrate in comparison with those of the reported antifouling coatings with lubricating effect. (c1) Water contact angles and (c2) surface energies (where γ denotes the total surface energy, and γ_s and γ_p the dispersion and polar components of the surface energy) of DA-PU, Schiff-PU, epoxy substrate, RILNs/epoxy substrate and NSCC/epoxy substrate. (d1) Fluorescence spectra of NSCC and RILNs collected under the excitation wavelengths of 300, 350, and 400 nm, respectively. (e1) Surface profiles of RILNs/epoxy substrate after immersion in acidic solution (pH = 3.0) for different times until the surface was smooth again (5 weeks). R_a represents the arithmetic average roughness. (e2) Raman spectra of RILNs/epoxy substrate after immersion in acidic solution (pH = 3.0) for different times until there was no further change (5 weeks). (e3) Water contact angles and (d5) surface energies of RILNs/epoxy substrate before and after immersion in acidic solution (pH = 3.0) for 5 weeks.

of our NSCC is superior to those of other reported in the literatures.

It is worth noting that when the surficial SLPU is completely washed away after long-term service in sea, the Schiff-PU rich layer of the stratified RILNs is exposed and plays the role of hydrophobic antifouling (Fig. 1j). Accordingly, its antifouling effect should also be known. Since the Schiff base in RILNs emits fluorescence [34], which would interfere with the fluorescent signals of any adhered protein BSA, the examination of the resistance to protein has to be skipped and we go straight to the evaluation of antifouling performance of RILNs by means of antibacterial tests that are more rigorous. As shown in Fig. 2c2, the

antibacterial rate under static conditions is 95.6 %, slightly lower than the value of NSCC but still at a very good level. The antibacterial habit of RILNs after the same seawater flushing is shown in Fig. 2d2. After 5 days, individual bacteria begin to adhere to the surface (antibacterial rate = 98.5 %), and more adhered bacteria appear after 7 days. Nevertheless, most areas of the RILNs still maintain good enough antifouling efficacy. Comparatively, considerable accumulation and stacking of the bacteria are found on the control in this case (Fig. 2d and 3).

The minor difference in antifouling performance between the RILNs and NSCC should originate from the difference between less-sticky

hydrophobic surface and non-sticky slippery hydrophobic surface, because both layers contain CAZ. In other words, the antifouling coating with low friction surface seems to be more promising for future development.

3.3. Antifouling mechanisms

To unveil the mechanisms involved in the antifouling behaviors described in the last section, we'll carry out the discussion according to the sequence of the coating from outside to inside as follows.

Fig. 3a1 shows that the sliding angle of NSCC is at a very low level (4.51°) as compared to the other slippery coatings ($2\sim 24^\circ$) [35–51], which means the excellent slippery effect. When contaminants approach to the NSCC surface, the lubricating surface of the outer SLPU makes them difficult to gain a foothold, thus providing the fouling resistance (Fig. 3a) [52]. The solid-state self-lubrication is powerful enough to continuously work for much longer time than the SLIPS strategy as the amount of the lubricating fluid stored in the SLIPS coating has to be rather limited. Even if the solid lubricity diminishes over time and the fouling organisms may migrate closer to the surface, they would be disrupted by the grafted CAZ, the latest generation antibiotic-based high efficiency antibacterial agent (Fig. 3b). Meanwhile, despite the loss of the semi-immobilized PDMS chains on the SLPU that reduces the lubricating effect, the hydrophobicity of SLPU surface (Figs. 3c and 1) would not be affected. The surface free energy of NSCC (24.19 mJ m^{-2} , Figs. 3c and 2) meets the criteria required for the best antifouling effect ($20\sim 30 \text{ mJ m}^{-2}$) [53]. Hence, the dead biofouling can be easily washed away by the seawater (Fig. 3c), keeping the NSCC surface free of deposits (Fig. 2c1 and 2d1).

After prolonged operation in submerged marine environment, the SLPU on NSCC may be entirely removed, but the exposed Schiff-PU rich layer of the stratified RILNs still exhibits outstanding antifouling property (Fig. 2c2 and 2d2) owing to the following actions.

- (i) The fluorescence of Schiff base bonds of the RILNs can be excited within the wavelength range from 300 nm to 400 nm (Fig. 3d1), i. e. the high-efficiency band of sunlight entering seawater ($300\sim 420 \text{ nm}$), which would keep biofouling away as the fluorescent organisms (e.g., corals) in nature (Fig. 3d) [54]. In contrast, NSCC does not have this effect (Figs. 3d and 1), because the superimposed SLPU itself cannot emit fluorescence, and UV light has rather limited penetration depth in the opaque SLPU.
- (ii) The hydrophobic Schiff-PU rich surface of the RILNs (Fig. 3c) is unfavorable for adsorption of bio-fouler. Its surface free energy (22.8 mJ m^{-2} , Fig. 3c1 and 3c2) also falls within the above-mentioned optimal range for antifouling, which renders the surrounding marine organisms susceptibility to departure from the RILNs surface, especially with the aid of flow of seawater (Fig. 3c).
- (iii) In case a few biofouling are successfully attached to the RILNs surface, they would be eliminated by the included antifouling agent CAZ (Fig. 3b).
- (iv) Supposing that the RILNs resist pollutants for long enough, the antifouling effect would inevitably decline and accretion of the biofouling carcass on the surface results in the formation of a weak acidic microenvironment [55]. Cleavage of the Schiff-base bonds on the surface of RILNs and the correlated decomposition of the macromolecules would occur under the circumstances [56], realizing surface self-renewal as verified by the simulation experiments (Fig. 3e). That is, when the RILNs coating is immersed in acidic water ($\text{pH} = 3.0$, which helps to accelerate the experiments in the simulated acidic microenvironment caused by bacteria metabolism), surface roughness of the RILNs coating initially increases and then recovers with a rise in the immersion time (Figs. 3e and 1). The Raman spectra in Figs. 3e and 2 clearly reveal that the above changes correspond to the variations in the

amount of Schiff-PU. The area ratio of the characteristic peak of Schiff-PU to that of DA-PU rapidly decreases and then tends to increase (Fig. S13). The initial decline is attributed to the degradation of Schiff-PU, while the subsequent recovery is due to that the gradual hydrolysis of the PBA soft segments [57] of the hydrophilic DA-PU in the absence of the protection of the interlocked hydrophobic Schiff-PU exposes the sub-layer RILNs as the new surface and results in a re-rise in the proportion of Schiff-PU. Because the degraded DA-PU aren't falling fast enough, the Schiff-PU/DA-PU ratio is not completely restored after five weeks (Fig. S13). Nevertheless, it explains the fact that when the coating surface becomes smooth again after staying in the acidic environment for long enough (Fig. 3e1), its chemical structure is almost the same as that of the original version (Fig. 3e2). Furthermore, although the water contact angle of the renewed coating surface is slightly reduced, it retains to be hydrophobic (Fig. 3e3) and the surface energy is still in the optimal range for antifouling (Figs. 3e and 4) [53]. As a result, the remaining coating can realize not only fouling-resistant by means of the intrinsic hydrophobicity (Fig. 3c) but also fouling-killing through the grafted fungicides (Fig. 3b). The antifouling effect would not vanish until Schiff-PU is completely removed. The passive acid-responsive self-degradation differs from the mechanism of self-polishing antifouling coatings that depends on active degradation of main chains or side chains resulting from hydrolysis of ester groups or the like. The former is more flexible as it adapts to the actual pollution situation. In summary, NSCC has temporal evolution antifouling ability, which not only endow it with a certain degree of intelligence, but also prolongs the service life and durability of the coating. As time progresses, the cost-effectiveness of NSCC will gradually become apparent when compared with traditional coatings.

3.4. Mechanical properties

From an application perspective, coatings should be firmly stuck to the substrates no matter what their uses are, so that adhesion and cohesion are of great importance.

According to the wetting rule for building up strong adhesion, the surface free energy of an adhesive should be lower than that of the substrate, and the closer the surface free energies, the better the compatibility. Because the RILNs undergoes self-stratification during the coating film formation (Fig. 1e), the microzone near the epoxy substrate is actually dominated by the DA-PU rich part (Fig. 1h). It is seen from Figs. 3c and 2 that the surface free energies of DA-PU, Schiff-PU and unstratified RILNs are all lower than that of the epoxy substrate, and the value of DA-PU is the closest to that of the epoxy substrate, indicating that the DA-PU rich part of the stratified RILNs would have the best wetting at the interface. It fits with our design for the coating in terms of thermodynamics.

When the slurry of the blended DA-PU and Schiff-PU is applied on the epoxy substrate under heating, the temporarily disconnected DA bonds and Schiff base bonds simultaneously favor (i) spreading of the low viscosity decrosslinked subnetworks on the substrate and (ii) stratification of the embryonic form of RILNs. Due to the attraction of the epoxy substrate, the fragmented DA-PU would further diffuse into its grinding marks that are created by sanding in advance, and hence mechanical interlocking between the DA-PU rich part of the RILNs and epoxy substrate is established upon the subsequent reconstruction of the reversible bonds in the course of cooling, accompanying with the formation of stratified RILNs.

As can be seen from the cross-sectional views of NSCC/epoxy substrate (Fig. S14), the interface is relatively flat in the case of unpolished epoxy substrate (Fig. S14a), while it becomes rather uneven when the epoxy substrate has been sandpapered forming larger contact area (Fig. S14b). By taking advantage of the mechanical interlocking, the

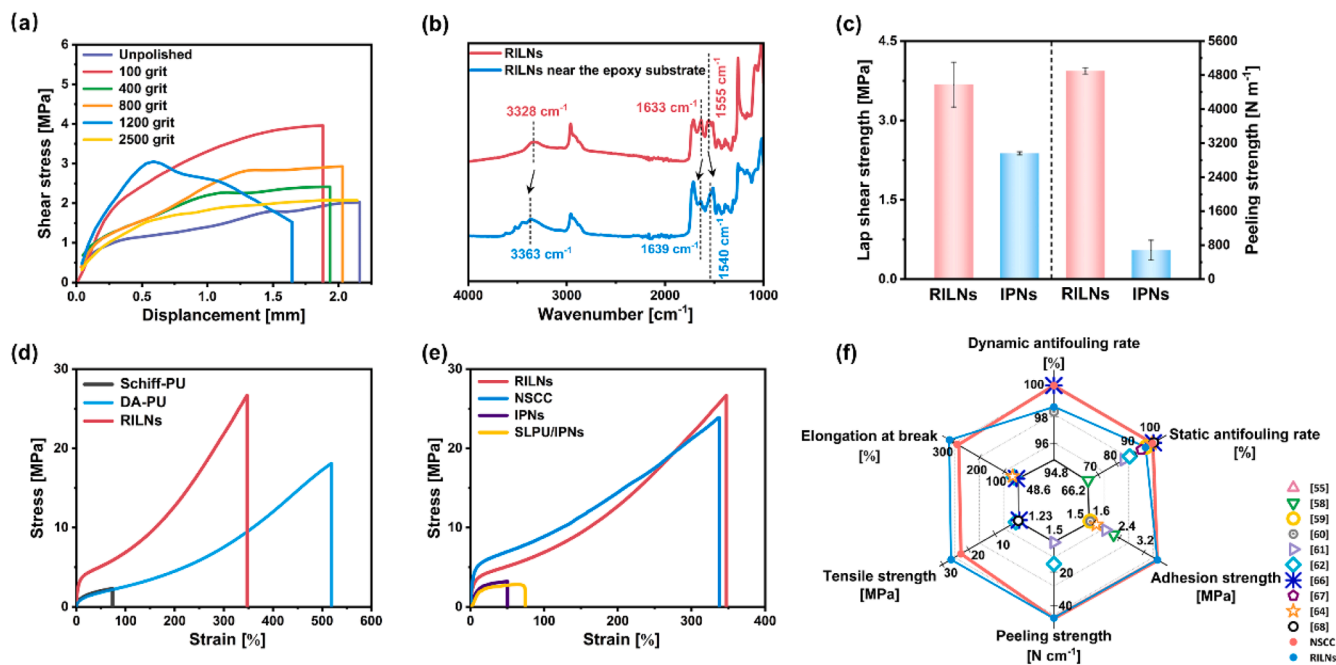


Fig. 4. Mechanical properties. (a) Shear stress versus displacement measured during the lap shear test of RILNs/epoxy substrates. The epoxy substrates were either unpolished or polished with sandpapers of different grit numbers (as shown in the legend) in advance. (b) FTIR spectra of RILNs and RILNs near epoxy substrate. Note: To obtain the RILNs near epoxy substrate for the measurement, the RILNs solution was applied to epoxy and dried. Then, the cured RILNs film was peeled off and the side near epoxy substrate was measured by FTIR in terms of ATR method. (c) Lap shear strength and peeling strength of RILNs in comparison with those of IPNs. (d) Typical tensile stress-strain curves of Schiff-PU, DA-PU and RILNs. (e) Typical tensile stress-strain curves of RILNs, IPNs, NSCC and SLPU/IPNs. (f) Dynamic/static antifouling rate, adhesion strength, peeling strength, tensile strength and elongation at break of NSCC and RILNs in comparison with other antifouling coatings reported in the literatures.

adhesion of the coating to the polished epoxy substrate is greatly strengthened, which is evidenced by comparing the shear strengths and peeling strengths of the systems with unpolished and polished substrates (Figs. 4a and S15). Because the sandpaper with lower grit number would generate coarser finish and stronger mechanical interlock, 100 grit sandpaper is used in most experiments here in this work to achieve the optimal adhesion effect, unless otherwise specified.

In addition, the interaction between the polarity groups of DA-PU (COOH, NH) (Fig. S3) and epoxy group of the epoxy substrate would generate hydrogen bonding, which also helps to raise the bonding of the coating to the substrate. As shown in Fig. 4b, the stretching vibration peak of NH of RILNs exhibits red shift (from 1555 cm^{-1} to 1540 cm^{-1}), and the absorptions of COOH (which appear at 3328 cm^{-1} and 1633 cm^{-1} due to stretching vibration of C = O and O—H, respectively) also shift to 3363 cm^{-1} and 1639 cm^{-1} after forming adhesion with the epoxy substrate. These prove the existence of hydrogen bonds between RILNs and epoxy substrate. Fig. 4c show the lap-shear strength (with average of 3.68 MPa) and peeling strength (with average of 4352.72 N m^{-1}) of RILNs/epoxy substrate joints, which are higher than the data of the other coatings on epoxy substrate (adhesion strengths: $1.5\sim 2.5\text{ MPa}$; peeling strengths: $150\sim 1500\text{ N m}^{-1}$) [58–64]. Because the results describe the capabilities of maintaining adhesion when force is applied parallel to the bond line (referring to lap-shear test) and perpendicular to the bond line (referring to 180° peeling test), respectively, it can be deduced that our coating has good affinity for the substrate even in the complex stress environment of the ocean.

Interestingly, Fig. 4c also illustrates the adhesion properties of simultaneous IPNs for comparison, which are synthesized from the same chemicals as the RILNs, except that no reversible covalent bonds are involved. The significantly poorer adhesion of the IPNs is attributed to the fact that (i) their inherent phase-separation (Fig. S16) hinders the intimate contact between the thermodynamically favorable components and the substrate, and (ii) their microstructure is unable to nimbly respond to the external conditions. Besides, the lap shear curves

(Fig. S17a) indicate that RILNs hold for longer and are much less vulnerable to be damaged than IPNs. In the case of peeling tests, both RILNs and IPNs are capable of stripping out of the entire bonding region from beginning to end (Fig. S17b). However, the curve of RILNs greatly fluctuates, whereas that of IPNs is much smoother, meaning that the adhesion formed by RILNs consumes more energy in the course of forced separation. The comparison shows the value of using RILNs in this work from another angle.

Besides strong adhesion to the substrate, the necessary durability of coatings also requires decent cohesion to avoid premature cohesive failure inside the coating body. As shown in Fig. 4d, tensile strength and toughness (estimated from the area under the tensile stress-strain curve) of the RILNs are 26.68 MPa and 41.51 MJ m^{-3} , respectively, reaching the level of commodity plastics. The synergy derived from the interlocked subnetworks (DA-PU and Schiff-PU) homogenizes the applied stress [65] and hence the RILNs possess much higher strength than the two subnetworks, which raises the resistance of the coating against the external force. Following the application of SLPU, the tensile properties (Fig. 4e) of NSCC remain highly robust, exhibiting no impact from SLPU. In contrast, the tensile properties of IPNs with or without superimposed SLPU are both markedly inferior due to the inherent phase-separation between the two components. Besides, the hardness of NSCC is determined to reach the highest level of 6H via the pencil test, which is sufficient to endure long-term use. The high hardness is attributed to the fact that the outer layer of NSCC is a solid slippery coating composed of highly covalently cross-linked polymers.

4. Conclusions

The contradictory non-sticky (towards marine organisms) and sticky (towards the protected marine infrastructures) properties required by antifouling coating are successfully united in the NSCC developed in the present work. In addition, temporally and spatially controllable multiple antifouling mechanisms are introduced into the NSCC for improving its

reliability and durability.

Specifically, hydrophobic and hydrophilic polyurethane networks containing reversible DA and Schiffbase bonds are combined at the molecular level by taking advantage of the specific interlocking architecture of RILNs and the associated forced miscibility. Based on the thermodynamically induced self-stratification of the RILNs, furthermore, the hydrophobic polyurethane-rich part is interpenetrated by a tailor-made low friction polyurethane outward layer, and the hydrophilic polyurethane-rich part is adhered to the epoxy substrate via mechanical interlocking and hydrogen bonds. As a result, a series of non-contact and contact antifouling strategies are seamlessly integrated with high cohesion and interfacial adhesion in the NSCC. The non-contact antifouling strategies include (i) solid-state slippery and hydrophobicity provided by PDMS, and (ii) fluorescence of Schiff base bonds, which keep fouling organisms away from the coating, while the contact ones involve (i) the use of the grafted CAZ for killing affixed microbes and (ii) surface detachment of the self-degraded polyurethane (caused by scission of Schiff base bonds) that carries away the accumulated biofouling.

The experimental results indicate that either the surface or the sub-surface of our NSCC exhibits excellent resistance to protein and bacteria under both static and dynamic conditions. The shortcomings of existing antifouling coatings, like easy depletion of liquid lubricant, low cohesion and adhesion, poor static antifouling effect, and uncontrollable hydrolysis of the self-polishing coating polymer, are thus overcome. Consequently, the properties of the NSCC and RILNs (i.e. NSCC excluding the slippery SLP outer covering) comprehensively outperform those of the analogues reported in the literatures (Fig. 4f) [55, 58–62, 64, 66–68].

The adaptively awakening of different combinations of antifouling effects arranged in different parts on the time dimension, coupled with the mechanical robustness, sets up a novel approach of coping with the destruction of seawater on antifouling coatings during long-term operation and greatly improving their service life. From the long-term point of view, more durable antifouling coatings can reduce the frequency of coating reapplication/renewal, thereby significantly enhancing the cost efficiency. Moreover, the structural design of RILNs, as well as its combination with solid slippery coating, can be adjusted in accordance with the actual circumstances, offering a highly scalable template for the development of antifouling coatings.

CRedit authorship contribution statement

Zheng Yue Wang: Writing – original draft, Investigation, Formal analysis, Data curation. **Yuyang Cao:** Investigation. **Lanyue Zhang:** Investigation. **Ze Ping Zhang:** Writing – review & editing, Funding acquisition. **Min Zhi Rong:** Writing – review & editing, Funding acquisition. **Ming Qiu Zhang:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.apmt.2025.102611](https://doi.org/10.1016/j.apmt.2025.102611).

Data availability

The data that support the findings of this study are available from the corresponding authors on request.

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